Estimating Tip Contact Forces for Concentric Tube Continuum Robots Based on Backbone Deflection

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***Abstract*—Concentric Tube Continuum Robots are among the smallest and most flexible instruments in development for minimally invasive surgery, thereby enabling operations in areas within the human body that are difficult to reach. Unfortunately, integrating state-of-the-art force sensors is challenging for these robots due to their small form factor, although contact forces are essential information in surgical procedures. In this work, we propose a novel data-driven approach based on Deep Direct Cascade Learning (DDCL) to create a virtual sensor for computing the tip contact force of Concentric Tube Continuum Robots. By exploiting the robot's backbone's inherent elasticity, deflection is used to estimate the respective external tip contact force. We evaluate our approach on different data representations for a single tube and apply it subsequently on a three-segment Concentric Tube Continuum Robot. Furthermore, we devise a novel transfer learning approach through DDCL to improve the estimation accuracy by pre-training a cascaded network with simulated data. Subsequently, we fine-tune the network based on a small real-world data set recorded from the physical robot.**

***Index Terms*—Concentric tube continuum robots, force estimation, neural networks, deep direct cascade network.**

I. Introduction

CONCENTRIC Tube Continuum Robots enable surgical procedures that are hard to conduct or even impossible with classical minimally invasive instruments due to their flexibility and small size. They are composed of multiple, superelastic, pre-curved concentric tubes made from the shape memory alloy Nitinol (NiTi). Over the last decade, they have been heavily investigated for the use in medical procedures. A grand challenge remains in the development of force sensing for a Concentric Tube Continuum Robot (CTCR) [1]. Because of their small dimensions, which often have an inner tube diameter less than 1 mm, there is little space to embed sensors additionally to tools such as grippers. However, the measurement of contact forces is essential information in surgical operations, e.g., to gain knowledge about the bio-mechanical characteristics of the tissue. A Force/Torque-sensor (FT-sensor) can be applied for measuring force for minimally invasive surgery. A tip force sensor based on the electrical resistance of fluid-filled channels was proposed by [2] and showed possible use for CTCRs. Another possible solution is fiber Bragg gratings (FBG) sensors, which enables real-time force and shape sensing and support integration into CTCRs [3]. However, adding sensors to a CTCR causes problems through the added weight, which possibly introduces additional movement constraints for the tubes and reduces space in the working channel for other tools. Most of the problems do not apply for FBG-sensing, but integrating such a sensor is challenging [4] and costly. In contrast, our approach measures force by indirectly using the robot itself as a sensor, utilizing its elasticity and shape-changes for intrinsic force sensing.

For continuum robots, intrinsic force sensing was introduced for robots with multiple backbones and multiple parts separated by disks in [5]. The tip contact wrench is calculated by the actuation loads, resulting in an average sensing error of 3.1% (3.335 $mN \pm 8.1423$ mN). Another intrinsic force sensing approach applied an Extended Kalman Filter (EKF) to estimate the tip forces by tracking the robot tip pose and actuation torques in [6]. However, in this work, the robot is restricted to a planar model, only bending in the x- and y-direction and it shows a high error when force is applied along the robot's backbone.

Promising results have also been shown for steerable catheters. In [7] and [8], the effect of contact forces on the deflection of an ablation catheter was investigated. A calculated dimensionless force index was introduced, based on the curvature of the catheter and measured forces, which were clustered by a Gaussian Mixture Model into three force ranges based on [9]. Experimental results showed a detection accuracy of 80%. Similar to [6], an iterative adaptive EKF was applied for a steerable catheter in [10]. A root mean square error (RMSE) of 1 ± 4.1 mN was achieved for a scenario simulating ablation forces. Additionally, for a scenario simulating an environment of intracardiac catherization, they achieved an RMSE of 2.8 ± 1.9 mN with the simulation of bloodflow and an RMSE of 3.5 ± 1.2 mN simulating beating heart dynamics. In [11], it was shown that it is possible to use the scaffolding of a robotic endoscope with segment-bending to increase tip contact forces and to estimate the contact force at three points along the robot by using tendon tension. A finite element model was created for the robot and forces at the rear, middle and front were estimated. Experiments showed an RMSE of 15.7 mN, 68 mN,

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and 103.4 *mN* respectively, each measured by a FT-sensor. In [12], a planar Cosserat rod model-based method was applied for a catheter to estimate the tip contact force magnitude and its direction with an accuracy of 89.50% (mean error 0.7 *mN*) and 88.13% (mean error 0.931°), respectively. In [13], a quasi-static model of an intracardiac catheter using a piece-wise circular arc approach was introduced and the force was estimated by using the tip pose. An accuracy of ±20 *mN* and ±10° was achieved for force magnitude and direction, respectively.

The referenced state-of-the-art approaches either simplified the problem by considering the robot as planar or exploited information of motor torques and tendon tension to infer internal forces. In contrast, our approach considers the full 3D workspace of the CTCR and, due to the external actuation unit, there is no direct relationship between torques and internal forces in the tube that could be exploited. Thus we follow a novel approach to create a virtual force-from-shape sensor through machine learning. To this aim, the present work introduces a novel data-driven approach using a deep cascade of Artificial Neural Networks (ANNs) to estimate contact forces.

The application of ANNs to learn models for a CTCR is a relatively new approach and has been done to the best of our knowledge only for learning kinematics models [14], [15], [16] and shape sensing [17]. In [14], a one-hidden-layer Multi-Layer Perceptron (MLP) learned the forward kinematics (RMSE 0.2 *mm*) and a two-hidden-layer MLP learned the inverse kinematics model (mean error of 0.8 *mm* and 0.1°) for a CTCR. A promising application of a neural network to learn the forward and inverse kinematics model is shown in [15] using recorded data from a CTCR and outperforming classical model-based approaches. The additional work [16] shows the importance of the representation of input and output data for the application of machine learning techniques to the learning of forward kinematics. A relative tip position error of 0.79% without load and 1.5% for a CTCR under load was achieved, outperforming the model-based approach described in [18]. In [17] the application of ANNs for shape estimation was investigated, showing an improved accuracy and computation time in comparison to physics-based models for CTCRs in free space. The approach achieved a maximum deviation from ground-truth data of 0.61 *mm*, an RMSE of 0.05 *mm*, and an accumulated error of 3.11 *mm* along the shaft.

Inspired by [6], we show how the elasticity of a CTCR can be used to estimate contact forces from its backbone deflection. We show how to obtain results comparable to the state-of-the-art in catheters by using a relatively simple supervised learning approach and then show how to improve the accuracy of the estimation to a level suitable for applications using Deep Direct Cascade Learning (DDCL). To achieve this, we also investigate how the representation of the backbone of the CTCR and its configuration, which serve as input data for the learning network, relates to the quality of the estimation. Finally, we show how transfer learning can be used in the deep cascade to tackle the problem when only a small training data set can be recorded from the real hardware. We pre-train a DDCL learner on simulated data and fine-tune the estimation towards the real robot by adding cascade layers trained only on an additional small real-world data set from that particular physical CTCR.

II. Methods

In this section, we present three types of learning algorithms that we evaluated for the use of virtual force sensing on the CTCR. We describe the different representations of its backbone and of the configuration of the CTCR. Furthermore, we describe the DDCL, a method to improve the approximation capability of the learning algorithms, which simultaneously simplifies the determination of the hyper-parameters of the applied algorithms. This method is called Direct Cascade Architecture (DCA), was first introduced in [19] and can be seen as a deep layered neural network for regression.

We aim to estimate the tip contact forces f_{ext} of a CTCR from its shape, that is its backbone deflection, by supervised learning. The shape can be described by the smooth curve $c \in \mathcal{C}$, where $\mathcal{C}$ is the shape-space containing all possible shapes of its backbone, which is physically determined by external forces and the superposition of the pre-curvatures of the robot's tubes, caused by a configuration $q \in \mathcal{Q}$, where $\mathcal{Q}$ is the possible configuration space of the robot's actuation unit. The curve c is described by the positions $r^*(s)$ along the backbone of the unloaded pre-curved tubes parameterized by the curve's arc length $s \in [0, 1]$. When tip contact forces f_{ext} are applied to the tip of the CTCR, a new curve is formed by $r(s)$ as a result of the deflection.

We use the model defined in [18] to determine the curve $c \in \mathcal{C}$ given $q \in \mathcal{Q}$ without and with external loads. Additionally, we use data recorded from the physical CTCR comprising $c \in \mathcal{C}$, $q \in \mathcal{Q}$, and f_{ext}.

Given a mathematical model for the deformation of a physical CTCR under load, we can compute $g : (f_{ext}, q) \rightarrow r(s)$. In this work, we aim to approximate the inverse $g^{-1} : (r(s), q) \rightarrow f_{ext}$ by applying supervised learning. It is well known in robotics that the particular representation of data, for instance of rotations and postures for such learning problems, can have crucial influence on the approximation quality of such learning, which was recently shown in [16]. Therefore, we also investigate empirically which representations of $\mathcal{C}$ and $\mathcal{Q}$ are suited for our problem.

A. Robot Configuration Space

A CTCR consists of $n \geq 2$ concentric superelastic NiTi-tubes nested in each other. These tubes can be thermally set to arbitrary shapes considering the recoverable strain limit of 5-8%. Each tube can be defined by several parameters such as length, diameter, and section curvature. For this work, we use a CTCR consisting of three tubes (see Fig. 1). Each tube $i \in \{1, 2, 3\}$ (1 being the innermost tube), consists of a straight and a curved section with respective lengths denoted by $L_{\mathrm{s},i}$ and $L_{\mathrm{c},i}$. The curved section is described by a constant curvature κ_i. Each tube has inner and outer diameters denoted by $d_{\mathrm{in},i}$ and $d_{\mathrm{out},i}$. The tube parameters for the robot used in this work are stated in Tab. I.

A CTCR is actuated by individually rotating and translating each tube. Rotation of each tube is denoted using the angle

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TABLE I
PARAMETERS OF THE CTCR USED IN THIS WORK

i	L_s	L_c	κ	d_{out}	d_{in}
1	359.0	41	28	0.5	0.4
2	202	100	12.4	0.9	0.7
3	100	70	4.37	1.5	1.2

All values in mm, except κ in m^{-1}

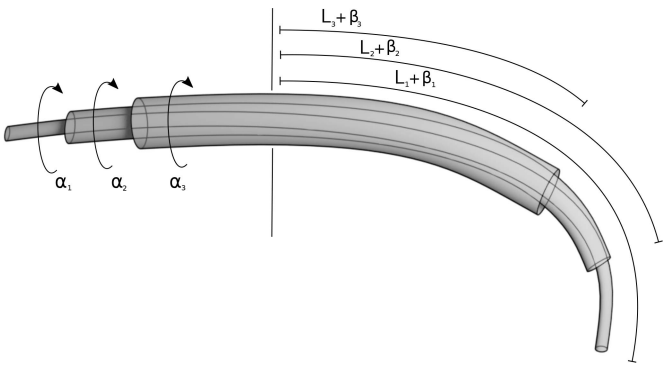

Fig. 1. A schematic of a three-tube CTCR with its configuration parameters.

(*rad*) $\alpha_i \in [-\pi, \pi)$. Each tube can be retracted up to its length (*mm*) $L_i = L_{s,i} + L_{c,i}$. Thus, tube translation is denoted by $\beta_i \in [-L_i, 0]$. The following mechanical limits of the robot have to be considered during translation of the tubes:

$$\beta_1 \leq \beta_2 \leq \beta_3 \leq \cdots \leq \beta_n \leq 0, \tag{1}$$

$$L_n + \beta_n \leq \cdots \leq L_1 + \beta_1. \tag{2}$$

A configuration q of the 6-DoF CTCR is described by

$$q = (\alpha_1, \beta_1, \alpha_2, \beta_2, \alpha_3, \beta_3). \tag{3}$$

For each configuration q, the physical shape of the CTCR will result in a curve c, which is represented by three-dimensional position vectors $r(s)$ along the tube's backbone and the orientation $R(s)$ at these positions. The end-effector of the CTCR is described by its position vector $r_{tip} = r(s = 1)$ and rotation matrix $R_{tip} = R(s = 1)$.

B. Data Representation

After obtaining the space curve $c \in \mathcal{C}$, the configuration $q \in \mathcal{Q}$, and the tip contact forces f_{ext}, different representations for input and output data are derived.

1) Data Representation of the Space Curve: Given the curve c of the backbone of a CTCR under load, a pose $P(s)$ for each position along it can be expressed by

$$P(s) = \begin{bmatrix} R(s) & r(s) \\ \mathbf{0}^T & 1 \end{bmatrix}. \tag{4}$$

The curve of each tube $i \in [1, 3]$ was discretized into 15 vertices $r_0^i, \ldots, r_{14}^i$, where $r_k^i \in \mathbb{R}^3$. The number of vertices was empirically estimated based on the average amount of possible retroreflective markers applied to a tube (see Section III-A1).

For each vertex r_k^i and r_{k+1}^i we calculated the connecting edge $e_k^i = r_k^i - r_{k-1}^i$ and the frames $\{t_k^i, n_k^i, b_k^i\}$:

$$\hat{t}_k^i = \frac{e_k^i}{|e_k^i|},$$
$$\hat{b}_k^i = \frac{\hat{t}_{k+1}^i \times \hat{t}_k^i}{|\hat{t}_{k+1}^i \times \hat{t}_k^i|},$$
$$\hat{n}_k^i = \hat{t}_k^i \times \hat{b}_k^i. \tag{5}$$

$$\bar{R}_k^i = \left[\hat{\boldsymbol{t}}_{\boldsymbol{k}}^{\boldsymbol{i}}, \hat{\boldsymbol{n}}_{\boldsymbol{k}}^{\boldsymbol{i}}, \hat{\boldsymbol{b}}_{\boldsymbol{k}}^{\boldsymbol{i}}\right], \tag{6}$$

where $\bar{R}_k^i$ is the discretized orientation $R(s) \in SO(3)$ on tube i. In the case that $\hat{t}_k^i = \hat{t}_{k+1}^i$, the frame is not defined. In this special case, we define $\bar{\kappa}_k^i = 0$ and $\bar{\tau}_k^i = 0$, where the bar indicates discretisation. The discrete point-wise curvature $\bar{\kappa}_k^i$ for a discretized curve $\bar{c} \in \bar{\mathcal{C}}$ is obtained via the turning angle ϕ_k^i as described in [20], [21].

$$\phi_k^i = \arccos\left(\hat{t}_{k-1}^i \cdot \hat{t}_k^i\right),$$
$$l_k^i = \frac{1}{2}\left(||e_{k-1}^i|| + ||e_k^i||\right),$$
$$\bar{\kappa}_k^i = \frac{2}{l_k^i} \tan\left(\frac{\phi_k^i}{2}\right), \tag{7}$$

where $l_0^1 = \frac{1}{2}(||e_0^1||)$ and $l_{14}^3 = \frac{1}{2}(||e_{13}^3||)$.

Additionally, the discrete torsion $\bar{\tau}_k^i$ of a curve $\bar{c} \in \bar{\mathcal{C}}$ can be computed by:

$$\bar{\tau}_k^i = \frac{2}{l_k^i} \arctan\left(\frac{\langle\, \hat{t}_k^i, \hat{b}_k^i \times \hat{b}_{k+1}^i \rangle}{1 + \langle\, \hat{b}_k^i, \hat{b}_{k+1}^i \rangle}\right) \tag{8}$$

We use $\bar{\kappa}_k^i$ and $\bar{\tau}_k^i$ as input data in this work.

2) Orientation Representation: In this work, we consider learning both only position information $r(s)$ of the curve points or full pose information $P(s)$ as input for the neural networks. In the latter case, finding the right representation [22] plays an important role in all aspects of robotic manipulation. This was already confirmed for forward kinematics learning of a CTCR in [16], where specific representations of the tip orientation in $SO(3)$ improved the accuracy of learning the forward kinematics of a CTCR. Thus for our discretized curve $\bar{c} \in \bar{\mathcal{C}}$ we calculated for each r_k^i an orientation frame $\bar{R}_k^i$ and evaluated two representations for $R(s)$: Roll-Pitch-Yaw angles and Quaternions.

a) Roll-pitch-yaw: Given the orientation matrix $R(s)$, where $r_{mn}(s)$ describes the component at row m and column n, the angles $\phi(s)$, $\theta(s)$, and $\gamma(s)$ representing roll, pitch and yaw can be extracted as described in [23]

$$\omega(s) = atan2(r_{23}(s), r_{33}(s))$$
$$\theta(s) = asin(r_{13}(s))$$
$$\gamma(s) = atan2(r_{12}(s), r_{11}(s)). \tag{9}$$

For the input to the ANN, the orientation in Roll-Pitch-Yaw is defined as $\Theta(s)^{rpy} = (\omega(s), \theta(s), \gamma(s))$.

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b) Unit quaternions: The unit quaternion representation of an orientation is defined as:

$$\psi = w + xi + yj + zk, \tag{10}$$

where Hamilton's rule is given by $i^2 = j^2 = k^2 = -1$ and $ijk = -1$, and $|\psi| = 1$. In comparison to Roll-Pitch-Yaw angles, unit quaternions avoid the possible problem of gimbal lock and represent a more compact representation than rotation matrices. The transformation function q_R stated in [23] was used to get $\Theta(s)^q = (w, x, y, z)^T$,

$$\Theta(s)^q = \frac{1}{2}\begin{bmatrix} (1 + r_{11}(s) + r_{22}(s) + r_{33}(s))^{\frac{1}{2}} \\ (r_{23}(s) - r_{32}(s))/(1 + r_{11}(s) + r_{22}(s) + r_{33}(s))^{\frac{1}{2}} \\ (r_{31}(s) - r_{12}(s))/(1 + r_{11}(s) + r_{22}(s) + r_{33}(s))^{\frac{1}{2}} \\ (r_{12}(s) - r_{21}(s))/(1 + r_{11}(s) + r_{22}(s) + r_{33}(s))^{\frac{1}{2}} \end{bmatrix}. \tag{11}$$

3) Force Representation: The external forces f_{ext} applied to the tip of a tube can be described as a vector $(f_x,\ f_y,\ f_z)$ in Cartesian space which is defined with reference to the tip frame $\bar{R}^1_{15}$ for one single tube and $\bar{R}^3_{15}$ for the CTCR. Note that the FT-sensor applied in this work is recording the forces in this representation.

4) Robot Configurations: An external actuation unit is used. The unit consists of three carriages each controlling the translation and rotation of one tube. As described in Section II-A the rotation of the i-th tube is described by α_i and the elongation by β_i.

a) Simple encoding: A simple form to represent the configuration q of the CTCR is given by the vector

$$q_s = (\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3), \tag{12}$$

where α_i is measured in *rad* and β_i in *mm*.

b) Cylindrical encoding: To examine a possible dependency of the accuracy on the representation of the configuration, we adopt the cylindrical representation proposed by [15]. With this, the configuration space is subdivided into two separate spaces: A revolute joint space A and a translational joint space B. While B contains the unchanged translational values of the simple form, A contains the angle α_i as $(\cos\alpha_i, \sin\alpha_i)$, which gives an unambiguous representation of the angle. The resulting feature vector for a three tube CTCR is given as:

$$q_c = (\zeta_1, \zeta_2, \zeta_3), \tag{13}$$

where $\zeta_i = (\cos\alpha_i, \sin\alpha_i, \beta_i)$. Rotation and translation is measured in the same units as q_s.

5) Input and Output Data: The final input data for training and testing consist of a combination of backbone description $\bar{\mathcal{C}}$, and configuration encoding $\mathcal{Q}$. The output data is given by a representation of f_{ext}.

C. Learning Algorithms

In this section, we describe the applied learning algorithms. We employ a supervised learning approach based on simulated and recorded real data. The prediction of contact forces for a CTCR renders a challenging task because of the high nonlinearity of the physical model (see Fig. 6).

1) Least Squares Linear Regression: The main goal of this work is to find a model-free approach to estimate contact forces for a CTCR by approximating the underlying function $g^{-1} : (r(s), \mathcal{Q}) \rightarrow f_{ext}$. Therefore, all learning algorithms are regression based, taking as input $\mathcal{C}$ or $(\mathcal{C}, \mathcal{Q})$ to estimate f_{ext}. As a baseline we apply Least Squares Linear Regression (LSLR) which represents a simple regression algorithm without any hyper-parameters.

2) Extreme Learning Machine: The Extreme Learning Machine (ELM) was introduced in [24] as a supervised learner. It is an ANN where each layer is connected all-to-all to the next and which projects input data to one higher dimensional hidden layer by means of fixed random connections W^{inp}. The output $\hat{y}$ for a given input X is calculated

$$\hat{y} = W^{out} \cdot \phi\big(W^{inp} \cdot X + b\big) = W^{out} \cdot H(X), \tag{14}$$

where the input weight matrix $W^{inp} \in \Gamma, \Gamma \sim U([-1, 1])$, W^{out} represents the modified output weight matrix, $b \in \mathbb{R}^h$ is the bias for the hidden neurons, where h is the dimension of the hidden layer and $H(X)$ describes the hidden layer neuron activation. We evaluate two common activation functions, the sigmoid activation function (SF) and Rectified Linear Unit (ReLu)

$$\phi(x)_{ReLu} = \begin{cases} x & \text{if } x > 0, \\ 0 & \text{otherwise} \end{cases}$$
$$\phi(x)_{sigmoid} = \frac{1}{1 + e^{-x}}. \tag{15}$$

A disadvantage of the ReLu is the local linear approximation, which leads to higher deviations when approximating smooth functions. Given the smooth model of the robot, we thus hypothesize that the sigmoidal activation function performs better and test this in our experiments. The use of random projections facilitates the use of ordinary least squares to obtain the weighted connections from hidden to the output layer.

The error has to be minimized by adjusting the weights in W^{out}:

$$E = \|\hat{y} - y\| = \|W^{out} \cdot H(X) - y\| \tag{16}$$

The solution for W^{out} can be found by applying

$$W^{out} = y \cdot H(X)^{\dagger}, \tag{17}$$

where $\dagger$ represents the pseudo-inverse operation, y a vector of target values, and $\hat{y}$ are the estimated values. Despite its simplicity, which benefits cascading as discussed below, the ELM is capable of universal function approximation, given a sufficient number of hidden layer neurons [24]. To improve the generalization performance, we use Ridge Regression instead of a simple pseudo-inverse to calculate the output weights of the ELM [25]. The output weights are calculated by:

$$W^{out} = \big(H(X)^T \cdot H(X) + \gamma I\big)^{-1} \cdot H(X)^T \cdot y, \tag{18}$$

where γ represents the regularization parameter and I the identity matrix. In this work, we refer to an ELM like this as Extreme Learning Machine with Ridge Regression (RRELM)

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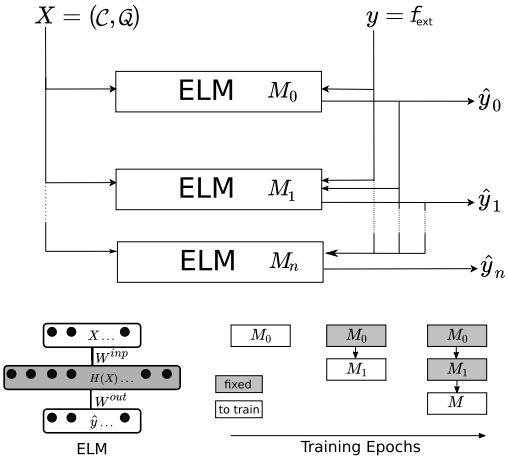

Fig. 2. General structure of the DCA-RRELM. After convergence, a new ELM is added and trained on the target y and the enhanced input X^i.

3) Multi-Layer Perceptrons: As a third learning algorithm, we employ a standard MLP. MLPs are ANNs with a simple feed-forward structure of several layers of neurons that are connected all-to-all, which is represented by their weight matrices. As shown in [26], MLPs with one hidden layer are capable of approximating any function given its data. Its forward calculation can be described by eq. (14). In contrast to the ELM, the weight matrix W_{inp} is adapted in training, which is done by gradient descent via back-propagation. Similar to the ELM, we compared the performance of the activation function $\phi(x)$ as ReLu and as SF. Furthermore, we applied a Ridge Regression also for the MLP. The regularization strength is regulated by the hyper-parameter γ_{MLP}. For training the MLP, we used stochastic gradient descent with a batch size of 200 and the Adam solver for weight optimization [27].

D. Deep Direct Cascade Learning

In this work, we employ an approach to efficiently learn supervised problems without the necessity of an intensive search of optimal hyper-parameters, e.g., the number of hidden layers in a network, called DDCL. DDCL is based to large parts on Direct Cascade Architectures introduced by [19]. It simplifies the optimization of hyper-parameters by incrementally stacking standalone learners as a deep cascade architecture (see Fig. 2).

A similar approach is the Deep Cascade Learning [28], which is an extension and adaption of Cascade Correlation learning [29]. In a common deep neural network architecture trained by back-propagation, all weights of all layers are trained at the same time. It changes the virtual inputs of the hidden layers constantly during training and thereby generates a moving target to approximate. Deep Cascade Learning achieves more efficient training through layerwise training and freezing of previously trained layers.

In contrast to Deep Cascade Learning, we do not train each layer by increasing the correlation of the virtual outputs of the last layer and the target, but by directly using the target data as the training target. The novelty of our approach is in the application of ELMs as cascade layers, combining the incremental optimization with a fast trainable ANN, and the way the data is utilized to this aim, which allows later to implement also transfer learning with mixed data sources from simulation and the real world.

The first DDCL learner (named M_0) is trained with the input data X^0 to reproduce the target data y. The learner trains until its error converges and stays fixed afterwards with no retraining. To improve the prediction accuracy, the algorithm introduces a new learner (M_1) as a new cascade to the architecture. To train the new learner, the previous cascade is activated with the training data X^0 to produce the prediction $\hat{y}^0$. Both the input X^0 and the output $\hat{y}^0$ of the first layer form the new input data vector $X^1 = (X^0, \hat{y}^0)$. The newly introduced learner is trained with input X^1 and target output y. New layers are added to the architecture until the global error converges, e.g., the decrease of the errors falls below a given threshold. For each cascade i, the input data set is $X^i = (X^0, \hat{y}^0, \ldots, \hat{y}^{i-1})$ and its prediction is given by $\hat{y}^i$. By introducing the previous prediction to the input, for each new cascade, the learning problem reduces to finding a model of the error of the previous architecture layers.

The Direct Cascade Architecture based on ANN achieves very good approximation results even for small data sets [19], which has inspired us to adapt this deep network approach for regression further to our needs. In particular, it is a great virtue of DDCL that they can use learners in the layers, which are not optimized by gradient decent like the ELM as we propose here and which combination we call Direct Cascade Architecture of ELMs with Ridge Regression (DCA-RRELM). This speeds up the learning process, reduces the number of hyper-parameters, and leverages global optimization per layer as shown in eq. (18). Further, we extend the application of DDCL to transfer learning, by pre-training multiple cascade layers with simulated data and then using recorded data to adapt the learned model to real measurements by adding additional cascades trained on the recorded data from the real robot.

III. Evaluation

We evaluate the performance of the previously described learning algorithms and different data representations for tip contact force estimation by deflection for a single tube and a CTCR with three concentric tubes. For the evaluation, a power set of all data representations given in Section II-B is built and the input data is normalized by Z-score normalization,

$$x' = \frac{x - \bar{x}}{\sigma_x}, \tag{19}$$

where x is the original feature vector, $\bar{x}$ its mean, and σ_x its variance. For comparing estimation performance, we use the RMSE. The LSLR, MLP, and ELM were written in Python using the scikit-learn [30] library. Training and testing were performed on a 64-bit Ubuntu Linux 16.04 operating system running on an Intel Core i7-6700HQ CPU at 2.60GHz.

A. Evaluation for a Single Tube for Single Layer Networks

To evaluate if the estimation of tip contact forces by deflection is feasible for CTCRs, we first investigate how well the

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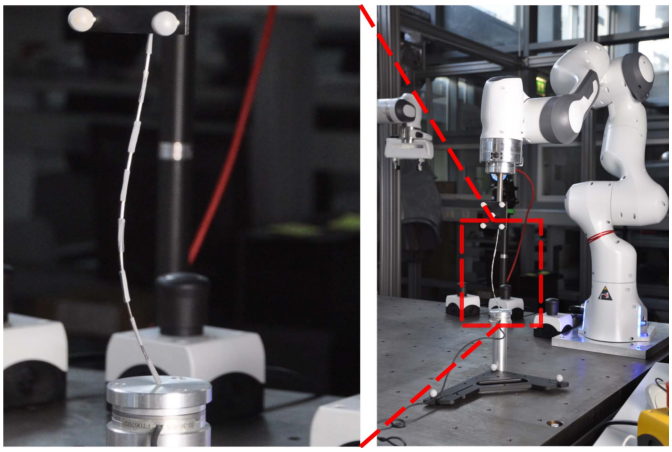

Fig. 3. A single NiTi-tube mounted as a tool of a Franka Emika Panda robot with retro-reflective markers and FT-sensor in front of the Optitrack motion capture system.

tip force can be estimated for a single NiTi-tube with a given pre-curvature.

1) Data Recording for a Single Tube: The evaluation for the single tube was conducted with recorded data from a physical NiTi-tube. We applied retro-reflective color to the tube, hereby creating point markers along its shaft. These markers were detected by six cameras of a motion capture system (Optitrack-Prime-13-System, NaturalPoint, USA) (see Fig. 9) with a recording frequency of 360 Hz and precision of less than 0.5 *mm*. Their positions in reference to the tracking system's base frame was transmitted via a ROS network to a central Linux system and stored together with the measured force at the FT-sensor in the same time frame. The forces were recorded by an ATI Nano 43 FT-sensor with a resolution of $1/128\ N$.

Note that the presented approach is independent of the type of tracking method. For in-vivo and ex-vivo applications, other tracking methods, e.g., [31] could be applied.

The tube was mounted on a flange of a Panda robot by Franka Emika with a custom drill chuck. A marker was attached to the end of the flange to mark the height of the drill chuck, which was used to later define the virtual unit base of the curve by the centre of the drill chuck. Seven markers with a length of 7 *mm* were attached to the 190 *mm* long test tube with a distance of 7 *mm* to each other. The tube had an outer diameter of 1 *mm* and an inner diameter of 0.9 *mm*. The tube consists of a 100 *mm* straight part and a 90 *mm* part with a default constant curvature of 0.004 m^{-1} in one direction without load or applied forces.

The setup is shown in Fig. 3. By moving the robot's flange with the attached tube perpendicular closer to the FT-sensor surface with a maximal movement of 10 *mm*, 15 *mm*, and 20 *mm*, we increased the force applied to the tip up to 2.29 *N*, thereby bending the test tube. The motion capture system was configured to be able to recognize non-spherical markers by adapting the filters. Although a position accuracy of 0.1 *mm* was achieved, up to four markers have not been recognized in some frames during the recording. Because of the noisy marker recognition, the recorded data was post-processed. For each sample, the curvature of the tube was interpolated by a B-spline using the recorded marker poses as control points to represent all tube shapes by equidistant points along the curve and for up-sampling the recording to a 15×1 vector. All curve samples were rotated and translated to align at the origin of a virtual unit base frame. For evaluation, we recorded the three movement distances towards the FT-sensor as separate data sets. Set 1 contained 15003 samples, set 2 contained 22501 samples, and set 3 contained 14003 samples. Each set was randomly split into 50% training samples and 50% test samples and all training and test sets were combined to the two data sets used in this evaluation. In total, we recorded 51507 data samples consisting of marker positions along the tube and force measurements of the FT-sensor.

TABLE II
PARAMETERS OF THE MLP EXCEPT h_{MLP} AND γ_{MLP} APPLIED IN THIS WORK

parameter	value
initial learning rate	0.001
maximum epochs	200
tolerance for optimization	$1 \cdot 10^{-4}$
β_1	0.9
β_2	0.999
ε	$1 \cdot 10^{-8}$

2) Parameter Optimization:

a) RRELM: The RRELM has only two hyper-parameters to optimize: the number of hidden layer neurons h_{RRELM} and the L2-regression parameter γ_{RRELM}. We optimized the parameter by applying a grid search over the intervals $h_{RRELM} \in [10, 2000]$ with a step size of 10 and $\gamma_{RRELM} \in [0.1, 1 \cdot 10^{-6}]$ with a step size of 0.0005. The grid search was performed on randomly selected samples from the training data including 30% of the whole sample set size. The result of the grid search determined $h_{RRELM} = 750$ and $\gamma_{RRELM} = 0.038$.

b) MLP: For the MLP, we optimized the hyper-parameters by grid search, too, with same range and step size and determined $h_{MLP} = 650$ and the L2-regression parameter $\gamma_{MLP} = 0.072$. Further parameters are given in the Tab. II

3) Results: The results for the single tube force estimation are summarized in Fig. 4 for forces represented as vectors in Cartesian Space. The tuples on the x-axis are representing the different encodings of the backbone space curve. The positions along the discretized space curve $\bar{r}$ are expressed by r, the curvature and torsion are given by κ and t, and the Roll-Pitch-Yaw and the Quaternion orientation are represented by e and q. The y-axis represents the RMSE in Newton.

As expected, the linear baseline LSLR performs worse than the other two learners and is not evaluated in greater detail.

For the RRELM and the MLP, the best results were achieved with an SF activation, as also expected (see Section II-C2). The best evaluation results for each learner are listed in Tab. III.

4) Discussion: The highest accuracy was achieved with an RRELM with SF with an RMSE of 113.74 *mN*, i.e., 5% w.r.t. a maximal force of 2.29 *N* and a combination of curvature and position information. It achieved a higher accuracy than

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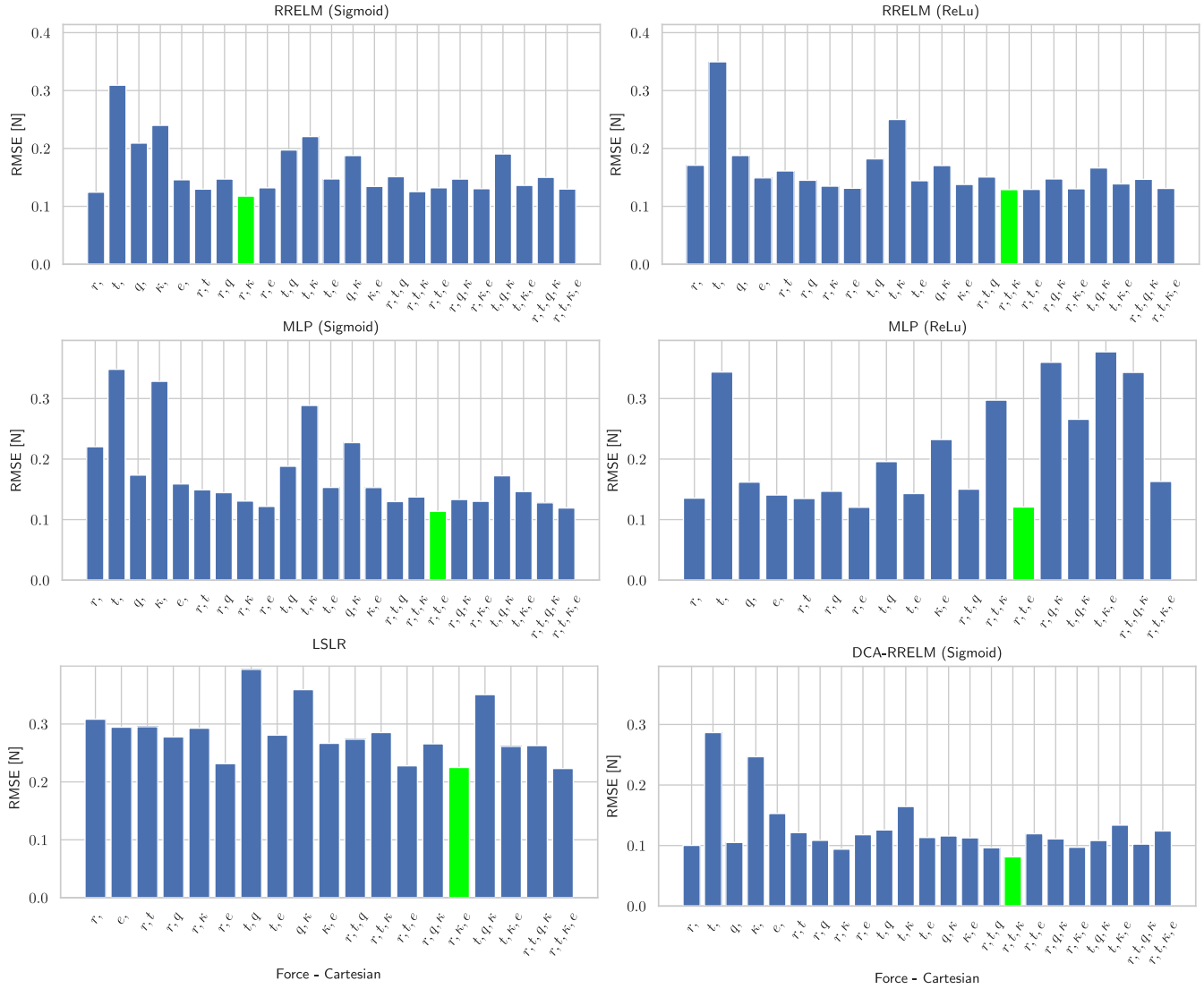

Fig. 4. Results for LSLR, ELM, and MLP with forces in Cartesian space. Only results with RSME $\leq 0.5\ N$ shown. The best results are marked in green, where r = positions along the backbone, κ = curvature, t = torsion, q = quaternion, and e = roll-pitch-yaw angles.

TABLE III
RESULTS OF LSLR, RRELM, AND MLP

Model	RMSE	Input Encodings
LSLR	223.09 mN	r, t, κ, e
RRELM (Sigmoid)	113.74 mN	r, κ
RRELM (ReLu)	128.87 mN	r, t, κ
MLP (Sigmoid)	117.66 mN	r, t, e
MLP (ReLu)	120.34 mN	r, e

the RRELM with ReLu, which can be explained considering the smoothness of the underlying model of the given data and given input data representations like curvature and torsion are highly non-linear transformations that are hard to approximate using a local linear model.

In general, learners who have only received torsion or curvature as input data have performed worse than the others. We explain this through the small changes in torsion and the negligible forces in x- and y-direction due to the experimental setup.

Interestingly, the combination with position data improves the performance. The better results can be explained by a strong underlying correlation between the local curvature and the position along the backbone. If tangents for two successive discrete positions along the backbone become parallel, the equations explained in Section II-B1 will result in zero curvature and torsion. Therefore, standalone curvature or torsion information can contain ambiguous samples, e.g., samples with only a few similar non-negative values. By augmenting the input data with information about the positions on the backbone, the curvature and torsion feature are coupled to local positions and thereby create a more descriptive feature set.

Furthermore, we conclude that the estimation of contact forces is possible based on the deflection for a tube. However, the performance of MLP and RRELM are not as good as the results for catheters in [10], [12], [13] and not in a range that would be sufficiently accurate for real application. Therefore, a deep learning approach (DCA-RRELM) was applied to this regression problem to improve the results.

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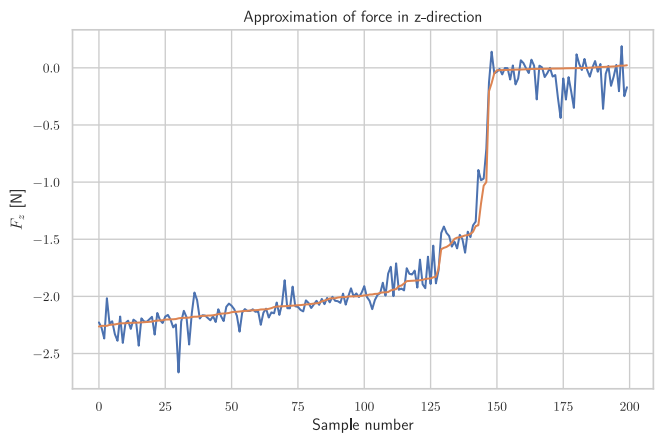

Fig. 5. A force profile given by the recorded real data (orange) and the approximation of the DCA-RRELM (blue).

B. Evaluation for a Single Tube With DDCL

We choose to improve our result by applying a DCA-RRELM with SF, described in Section II-D, for estimating the tip contact forces for a single tube of CTCR, because of the simplified tuning (only h_{RRELM} for the number of nodes in a layer and γ_{RRELM} for the regularization) in comparison to the MLP. The hyper-parameters for the DCA-RRELM has been set to the same values as for the simple single layer RRELM. Note, that due to its incremental approach of DDCL, the size of the hidden layer of each shallow network cascade is not important. Each additional cascaded layer reduces the error of the previous by implicitly learning the error model of it. To show this advantage, we trained an additional DCA-RRELM, with a hidden layer size of only 325 neurons. The input representation was chosen based on the best results achieved with the DCA-RRELM with h_{RRELM} and γ_{RRELM} as hyper-parameters. 5% of the training data was taken to form a validation set. With it, a stop condition was formulated, which was used to check network convergence. Convergence is defined as $0.001 \geq |e^{i}_{RRELM} - e^{i-1}_{RRELM}|$, that is, if the decrease of the error between successive layers of the cascade falls below the threshold.

1) Results: The best result was achieved by a combination of position data r, with κ and τ information as input, with an RMSE of 81.429 mN, i.e., 3.6% w.r.t. a maximal force of 2.29 N , and a cascade depth of 3. A similar result of an RMSE of 110.492 mN was achieved by the DCA-RRELM with only 325 hidden neurons per cascade with a total of 7 cascades. All results for the DCA-RRELM are shown in Fig. 4. Furthermore, the application of the DCA-RRELM on a test force profile from real measurements is shown in Fig. 5. For this application, the tube's tip was positioned close to the FT-sensor's contact area and the robot's flange was moved towards the sensor stopping 170 mm above the contact area. From that start position, the flange was moved slowly away from the sensor until no force was measured.

2) Discussion: The results shows an improvement thanks to the introduction of additional layers. Furthermore, the results of a DCA-RRELM with only 325 hidden neurons per cascade indicate that the DDCL approach reduces the importance of hyper-parameter fine-tuning. Although, a better result was achieved with the hyper-parameters taken from the grid search, for DDCL, it is not necessary to use optimal parameters, which reduces optimization and training time.

Like in the previous work [19], we do not observe overfitting despite the relatively small training data set, while exceeding the performance of a single layer learner. This is of high importance as we have to operate with a considerably smaller data set as, e.g., [15], [16], [17], given that real world measurements from the CTCR are difficult and tedious to obtain.

The results are comparable to those of the cited work and, considering the fact that only the deflection of the tube itself and no further information is used, indicates a possible use as a means of force estimation.

C. Evaluation for 3-Tube CTCR Based on Simulated Data

For further evaluation of the scalability of our DDCL approach, we trained the learners on simulated data for a three tube CTCR described in Section II-A. Additionally, we trained an MLP with multiple layers for comparison. For the tubes configuration, we generated two different representations for the joint space, which are described in Section II-B4. In contrast to the single tube evaluation, the data set generated for the three tube CTCR contains a greater variety of curves, which requires a more thorough sampling of the workspace. We applied a five-fold cross-validation for testing of the tip-forces estimation of the three tubes CTCR and measured the performance by the mean RMSE and the standard deviation σ for all five runs.

1) Data Generation in Simulation for CTCR: To generate data, we are using a C++-implementation of [18] to describe the kinematic behavior of CTCR given external loads. This model considers the robot's backbone to behave like an elastic rod and integrates all loads acting on it. It uses an adapted formulation of the Cosserat rod theory to derive a set of ordinary differential equations for the kinematic variables such as rod position and orientation. Due to the non-linearity of the resulting equations, a numerical solver is employed to determine these kinematic variables. Our implementation of this model receives a configuration q_s and a force vector $(f_x, f_y, f_z) \in \mathbb{R}^3$ and calculates discrete positions $\bar{r}_k \in \mathbb{R}^{1x3}$ where $k \in [0, 200)$, which describe the robot as a discretized curve $\bar{c} \in \bar{C}$. We chose 200 positions as a compromise between model accuracy and exhaustive calculation time with larger numbers of positions. As stated in [18], this model can achieve an accuracy in the range of 1.5–3% with respect to the robot's total length. The obtained data was post-processed by the same spline interpolation technique described above to fit the real data. We generated three data sets of 50,000 data samples, each with a maximum force magnitude of 0.1 N in x-, y-, and z-direction. Due to the very small diameter of the applied tubes in this work (the inner tube has only a diameter of 0.5 mm), larger forces are not applicable to every configuration in every angle and can lead to no numerical solution. Therefore, to generate an evenly sampled set, we restricted the force magnitude to 0.1 N.

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TABLE IV
RESULTS FOR DCA-RRELM AND MLP WITH TWO HIDDEN LAYERS

Data Set	Configuration	Model	RMSE (mN)	% wrt. max. force	Input Encodings
Full Workspace	q_s	DCA-RRELM	24.96 ± 1.756 mN	24,27 %	κ, r
	q_c	DCA-RRELM	24.266 ± 2.173 mN	24.27 %	κ, r
	q_s	MLP	$26.073 \pm 0,316$ mN	26.07 %	r, e
	q_c	MLP	26.162 ± 0.288 mN	26.16 %	t, κ, r
$\alpha_i \leq \pm 90°$	q_s	DCA-RRELM	8.482 ± 0.321 mN	8.48 %	κ, e, r
	q_c	DCA-RRELM	8.471 ± 1.389 mN	8.47 %	t, e, r
	q_s	MLP	9.799 ± 1.337 mN	9.8 %	κ, e, r
	q_c	MLP	9.85 ± 0.233 mN	9.85 %	t, e, r
$0.05m < \beta_i \leq 0$	q_s	DCA-RRELM	9.957 ± 1.387 mN	9.96 %	t, e, r
	q_c	DCA-RRELM	9.985 ± 1.571 mN	9.99 %	κ, e, r
	q_s	MLP	11.309 ± 0.426 mN	11.3 %	κ, e, r
	q_c	MLP	11.509 ± 0.53 mN	11.5 %	t, e, r

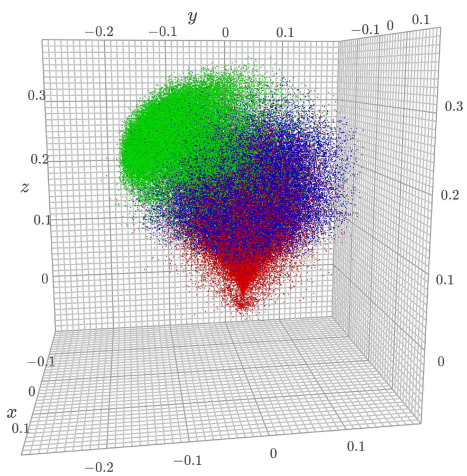

Fig. 6. The three sample sets are represented by the tip positions of the CTCR. Full workspace with applied forces (blue), restricted elongation (red), with $-90° < \alpha_i \leq 90°$ (green). ($x-$, $y-$, $z-$axes in m).

We generated three data sets of differently restricted workspaces represented in Fig. 6 by the generated tip positions: one without restrictions to the robot's configuration, one with limited elongation by $-0.05\ m < \beta_i \leq 0\ m$ and one with limited rotation by $-90° \leq \alpha_i \leq 90°$.

2) Parameter Optimization:

a) DCA-RRELM: We optimized the hyper-parameters for a single cascade layer by a grid search over the intervals $h_{DRC} \in [10, 2000]$ with step size 10 and $\gamma_{DRC} \in [0.1, 1 \cdot 10^{-6}]$ with step size 0.0005, where DRC stands for DCA-RRELM trained on the CTCR data. The result of the grid search determined $h_{DRC} = 600$ and $\gamma_{DRC} = 0.068$.

b) MLP: For the MLP, we determined an optimal layer count of two, testing networks between one to four layers, $h_{2MC} = 650$ and $\gamma_{2MC} = 0.072$, where 2MC stands for two layer MLP trained on the CTCR data. Further parameters are given in the Tab. II.

3) Results: Fig. 7 shows the results for the three described sample sets. For better readability, combinations with a $\sigma > 0.02$ for the RMSE for all runs have been omitted from the plot. The best results given for the targets in the full workspace are achieved for the combination of position data, torsion, and curvature as input for the network.

The DCA-RRELM achieved the best results given $\alpha_i \leq \pm 90°$ and q_c with an RMSE of 8.47 ± 1.389 mN. The best result for the MLP learner was also achieved given restricted α_i and using q_s. The RMSE was 9.799 ± 1.337 mN. The best results for all data sets are shown in Tab. IV.

4) Discussion: As with the single tube experiment, we were able to see an improvement in accuracy for the combination of curvature, torsion and position data in the input. Additionally, including orientation information in the input data improved the accuracy slightly. In contrast to the evaluation with a single tube, the information about curvature or torsion yields better accuracy, which can be explained by the greater variety of curve shapes given by the superposition of several tubes. The greatest improvement in accuracy is achieved by including the position information into the input data. As discussed above, this can be explained by the local correlation of the position with the curvature and torsion.

As shown, the learners were able to estimate the force but only with limited accuracy for sampling from the whole workspace. A reason for this could be an under-sampling given the complex nature of the workspace of the CTCR. An efficient way to sample from this space is still an open question.

The DCA-RRELM showed the best results in our comparison for estimating forces considering the whole workspace of the CTCR. It adapts better to the data because each additional layer only learns the error model of the previous stack. In contrast to the MLP, the variance in RMSE for each run is higher for the DCA-RRELM, which can be explained by the randomly drawn input weights of the RRELM, which stay fixed and can lead to an unfavourable hidden state. The results show that an estimation of the tip contact forces is possible, and even results in a high accuracy using a restricted workspace. Considering that CTCR tubes are custom designed for each procedure individually using computational optimization [32], [33], [34], [35], the reduction of workspace complexity as given in our evaluation is not a restriction in terms of potential applications.

We saw a similarly improved estimation accuracy, as shown in [15], [16] when using q_c as input data.

D. Evaluation for a CTCR Based on Recorded Real Data

We recorded a small data set of 349 samples from a real CTCR with the given tube parameters described in Section II-A. In addition to a simple supervised approach, due to the relatively small data set, we evaluated the accuracy of a network pre-trained with simulated data and the

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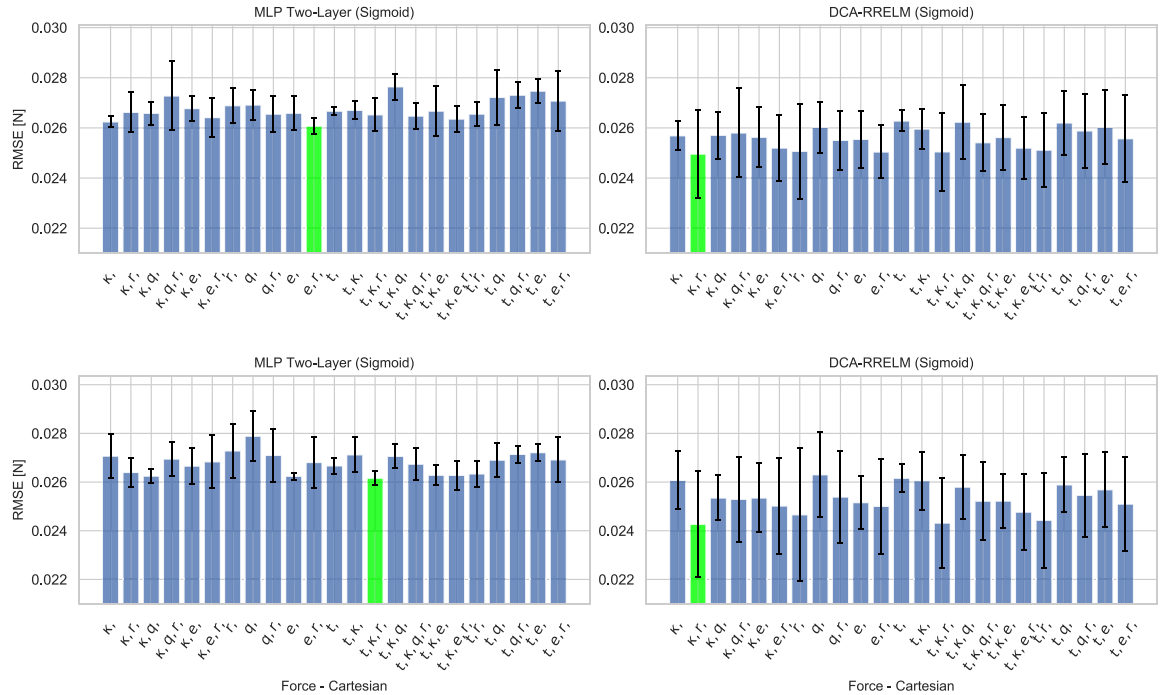
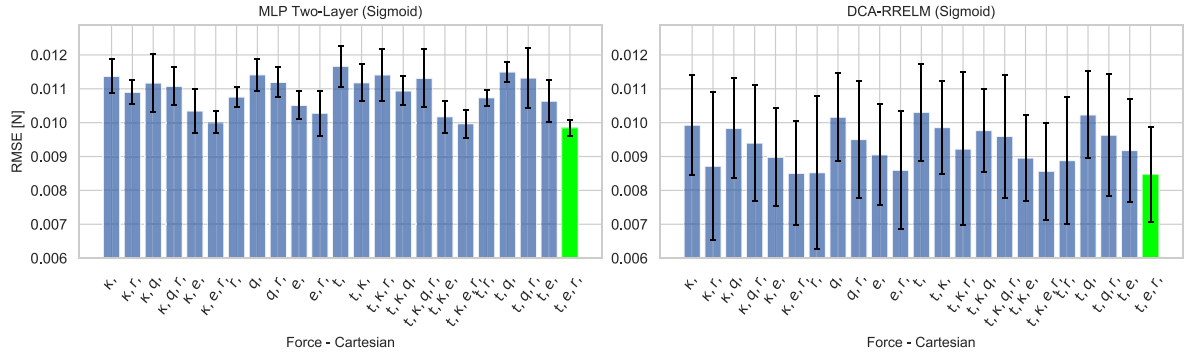
(a) samples from entire workspace with simple configuration (upper plots) and cylindrical (lower plots) configuration



(b) $\alpha_i \leq \pm 90°$ with cylindrical configuration



(c) $0.05\ m < \beta_i \leq 0\ m$ with cylindrical configuration

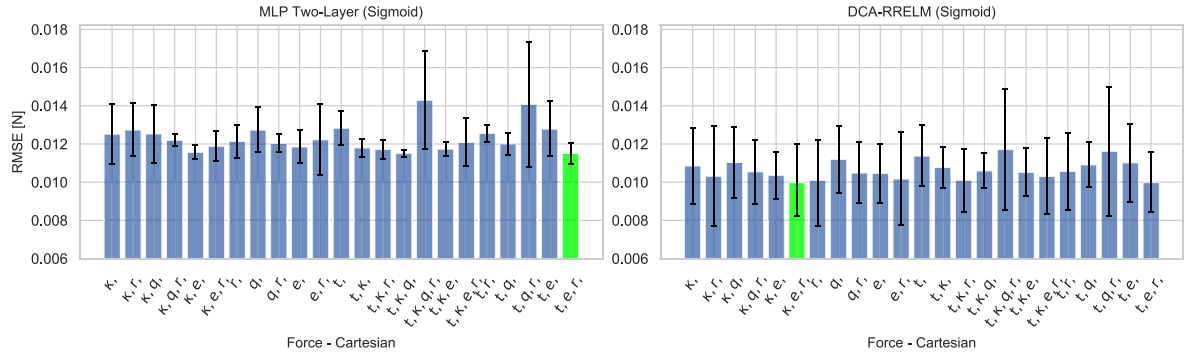


Fig. 7. RSME for the five-fold cross-validation with σ for DCA-RRELM and a two-layer MLP with simple and cylindrical configuration representation for samples from the entire workspace, workspace restricted to $\alpha_i = \pm 90°$, and $0.05\ m < \beta_i \leq 0\ m$ for external forces $F < 0.1\ N$. Where r = positions along the backbone, κ = curvature, t = torsion, q = quaternion, and e = roll-pitch-yaw angles. Best results are highlighted green. Results with $\sigma \geq 0.02$ have been omitted.

performance of an inductive transfer learning approach similar to the approaches explained in [36], [37]. In contrast to the pre-trained case, where the network was trained with the simulated data and tested on the real recordings, the transfer learning approach splits the recording data into an equal part of training and testing data. The training data is then used to adapt

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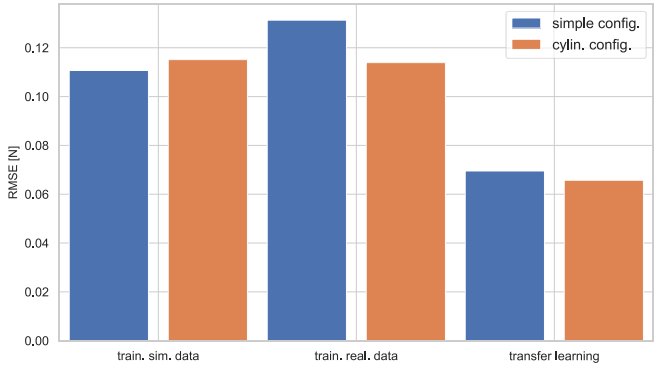

Fig. 8. RSME for DCA-RRELM with simple and cylindrical configuration representation, comparing training on real recordings only, with a pre-trained network, and with transfer learning.

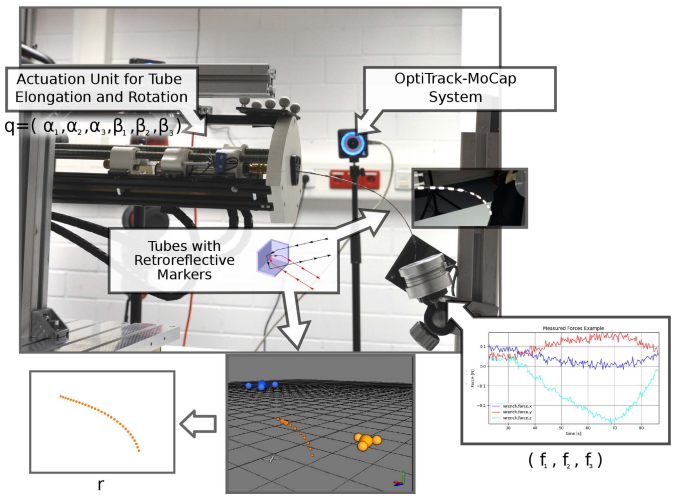

Fig. 9. CTCR with retro-reflective markers and FT-sensor in front of the Optitrack motion capture system.

the pre-trained learner to the real recordings. By training with real data, additional layers are added to the cascade, learning the difference between simulated and real data.

1) Recording Data for CTCR: For the recording of three concentric combined tubes of the described CTCR, we used 12 markers of size 5 *mm* with an equal distance of 5 *mm*. We recorded 349 data samples consisting of samples without (20%) and with applied forces (80%). Forces were generated by driving the inner tube orthogonally against the FT-sensor. A B-spline interpolation was applied and 45 positions spaced evenly along the generated curve were computed. In contrast to the single tube scenario, a relocation of the FT-sensor was necessary to increase the recording workspace. To record the position and orientation of the FT-sensor, we attached a marker to the side of the FT-sensor as well. Furthermore, the pose of the actuation unit was tracked, too, to calculate the virtual base of the curve of the tube and to apply the same transformation as given in the single tube scenario. The recorded data had a maximum force magnitude of $f_{ext} \leq 390$ *mN*. Pre-training of the networks was based on an additional data set of 50,000 samples with the same maximum force magnitudes and the same restriction to angular movement as the recorded data from the real robot.

2) Results: The results of our evaluation for recorded real data are shown in Fig. 8. The DCA-RRELM achieved an RMSE of 130 *mN* and 110 *mN* for simple and cylindrical configuration for training and testing only on the real data. For each training run, the DCA-RRELM only created two or three layers until convergence. The highest accuracy for a pre-trained network was an RMSE of 110 *mN* given a simple configuration encoding and 120 *mN* for the cylindrical encoding. Transfer learning achieved an RMSE of 70 *mN* and 60 *mN* for simple and cylindrical configuration.

3) Discussion: Given the minimal data set, it is not surprising that the learners, which were trained only on the real recordings, performed poorly. Evaluating the data on a DCA-RRELM, which was pre-trained with simulation data, improves the performance, but without a significant difference. One reason for this can be explained by the error of the numerical model. It is necessary to compensate the error with further learning. This can be seen very well in the results of the transfer learning approach. The DCA-RRELM algorithm added four additional layers and reduced the error significantly. Hereby, we could show that this approach can help to optimize learners for the application on real robots.

IV. Conclusion

In this paper, we proposed a novel approach to estimate tip contact forces for CTCRs through supervised learning, considering only the backbone deflection and the configuration of the drive unit. Furthermore, we introduced DDCL, which enables efficient training without exhaustive hyper-parameter optimization by avoiding the moving target problem during the training of deep neural networks. Additionally, by employing ELMs as cascade layers, we reduced the training procedure to finding the inverse of one weight connection matrix. We investigated the impact of different data representation for the approximation accuracy and validate our technique by applying it to recorded data for a single NiTi-tube and simulated data for a three-tube CTCR. Finally, we showed a novel transfer learning approach based on DDCL, which enhances the usefulness of small recorded data sets by pre-training Deep Direct Cascade Network with data generated by a simulated model of the same robot. In contrast to model-based methods, errors due to unmodeled properties can be reduced by transfer learning.

The results of our work raise new open questions. Further investigations about the applicability of the transfer learning approach are necessary, given that we used a single small data set in this work. We achieved the best result of 8.471 ± 1.389 *mN* for tip contact force estimation for a three-tube CTCR by restricting the rotation of each tube to $\pm 90°$. Although a CTCR is utilized in a task space smaller than its full workspace, further investigations in an optimal sampling strategy have to be conducted to improve the estimation accuracy of our approach without restrictions. Additionally, we want to improve the recording technique for real robots.

Moreover, an investigation should be conducted regarding the application on the real robot in combination with control methods. They could enable the robot to comply with force

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limits in surgical procedures to automatically prevent involuntary puncturing of tissue. Potentially, it could also be possible to create force feedback for the surgeon and thereby allow safer interaction with the tissue.

While these questions shall be further investigated, we think that overall the results support the claim that force estimation with ANN by utilizing the backbone deflection of CTCRs has the potential to become a cost-reducing and easy to integrate method. This virtual sensing of forces-from-shape could enable safer applications of CTCRs without restricting tool usage through additional, space-demanding force sensors.

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